MCP: one way to plug tools in

The Model Context Protocol, so every tool doesn't need its own integration with every AI app.

60 min | Recorded | Class 11 of 15

11

What this session covers

01
The problem MCP fixes
2 slides

02
What a server offers
2 slides

03
Using it safely
2 slides

04
Code walkthrough
4 slides

© 2026 Devansh Parapalli

Section 1 of 4

The problem MCP fixes

Every app times every tool.

01

Without a standard, every AI app needs its own connector to every system

02

Three parts, like a plug and a socket

© 2026 Devansh Parapalli

Without a standard, every AI app needs its own connector to every system

Before

- 3 AI apps × 5 of our systems = 15 integrations
- Each built differently, each maintained separately
- A Tracklane change breaks all of them

With MCP

- Each system gets one MCP server
- Each app speaks MCP once
- 3 + 5 = 8 pieces, and they all fit

© 2026 Devansh Parapalli

Three parts, like a plug and a socket

The host is the app people use; the server is what we build.

01 Host
The AI app: a chat app, a code editor, our own agent. It runs the model.

02 Client
Lives inside the host. One per server it connects to.

03 Server
Ours. Exposes Tracklane lookups, runbooks, prompts. Knows nothing about which model is asking.

© 2026 Devansh Parapalli

Section 2 of 4

What a server offers

Tools, resources and prompts.

01

A server can offer three kinds of thing

02

Two ways to connect: on the same machine, or over the network

© 2026 Devansh Parapalli

A server can offer three kinds of thing

Tools	Actions the model can ask for: look up a shipment, check a batch. Same idea as class 10.
Resources	Things to read: a runbook, a document, a record. The host decides when to include them.
Prompts	Ready-made instructions a person picks from a menu: “triage this ticket”.

© 2026 Devansh Parapalli

Two ways to connect: on the same machine, or over the network

Local (stdio)
The host starts the server as a program on your laptop. Simple, private, one user.

Remote (HTTP)
The server runs somewhere central. Shared by a team, needs sign-in and access rules.

© 2026 Devansh Parapalli

Section 3 of 4

Using it safely

A server is code you run.

01

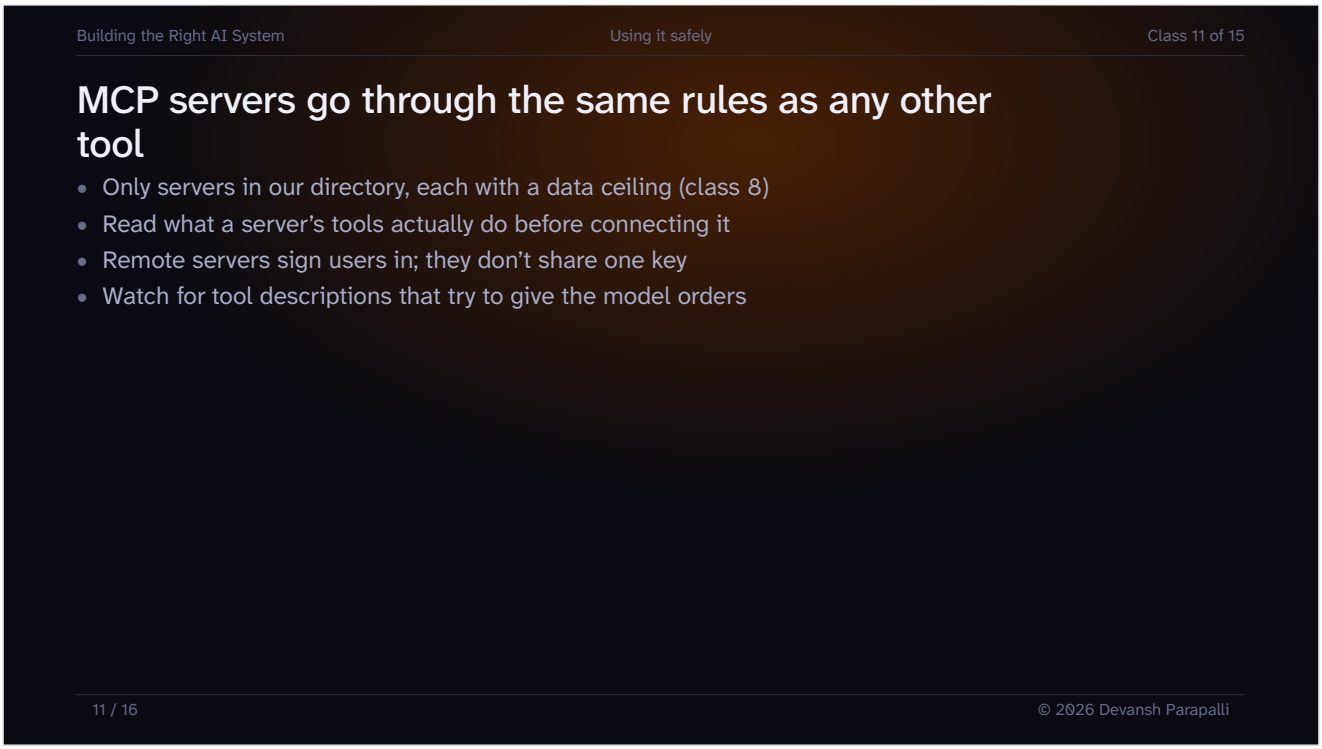
MCP servers go through the same rules as any other tool

© 2026 Devansh Parapalli

A server is **code you run**, so treat it that way.

Installing an MCP server is installing software. It can read what it's allowed to, and do what it's allowed to.

© 2026 Devansh Parapalli



A malicious server can put instructions in a tool's description. The model reads descriptions as trusted. That's why the directory exists.



One tool, one resource and one prompt in about twenty lines

code/class11/server.py

```python
mcp = FastMCP("lol-ops")

@mcp.tool()
def get_shipment(shipment_id: str) -> dict:
    """Where a LOL shipment is and why, e.g. LOL-SH-4472190."""
    return SHIPMENTS.get(shipment_id, {"error": "no such shipment"})

@mcp.resource("runbook://{name}")
def runbook(name: str) -> str:
    return (RUNBOOKS / f"{name}.md").read_text(encoding="utf-8")

@mcp.prompt()
def triage(ticket: str) -> str:
    return f"Route this LOL ticket to one queue and say why in one sentence:\n\n{ticket}"
```


© 2026 Devansh Parapalli

Adding it to a host is a few lines of config

code/class11/host-config.json

```json
{
  "mcpServers": {
    "lol-ops": { "command": "uv", "args": ["run", "code/class11/server.py"] }
  }
}
```


© 2026 Devansh Parapalli

Where today sits on the card

1. A script	The steps are known and don't change.
2. A rule	The decision is known and must be explainable line by line.
3. Classical ML	Labelled history, and the input is a table or short text.
4. A small trained model	Lots of text, images or audio, enough labels, cost per call matters.
5. An LLM with a prompt	Varied language in and out, few labels, someone checks the output.
6. An LLM plus retrieval, tools or agents	**Needs private knowledge, has to act, or takes several steps.**
7. Not yet	No ground truth, needs a guarantee, or nobody owns the result.

© 2026 Devansh Parapalli

What to do next

1. Connect `server.py` to an MCP host you're allowed to use and ask about LOL-SH-4472190.
2. Add a resource for the handwritten tickets, then decide whether it should exist.
3. Read the class 12 pre-reading on agents.

Questions after class: the course channel.

© 2026 Devansh Parapalli